

Viktor Trilar

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EDUCATION

Bachelor of Engineering — Information and Communications Technology Aug 2024 – Present
Turku University of Applied Sciences | Major: Data Engineering & AI | GPA 4.4 / 5.00

EXPERIENCE

Research Intern Apr – Jul 2026
National Institute of Technology, Sendai College — Japan | Supervisor: Prof. Jun Suzuki

- Built an automated system that checks whether students have correctly wired electronics breadboards from a phone camera photo, replacing manual teacher inspection
- Generated synthetic training images in Blender and trained AI models to detect wires, holes, and components (LEDs, resistors, transistors) on a breadboard with 99.5% accuracy
- Delivered a full web application for instructors to upload data, receive student submissions, and view class-wide analytics in Japanese/English

Applied AI Intern 2025 – Present
Cognitive Technologies Research Group (CoRe) — Turku, Finland

- Building AI systems for predicting industrial machine failures and inspecting manufactured parts for dimensional defects, running large-scale experiments on Finland's national supercomputer

PROJECTS

Predictive Maintenance — Sensor Anomaly Detection 2026
PyTorch · Transformers · Scikit-Learn · CSC Puhti HPC

- Generated synthetic data and trained a model that detects early signs of machine failure from sensor data without needing historical failure examples, covering 27 machine types; achieved AUC 0.897 across unseen machines

3D Part Inspection Pipeline 2026
Python · PyTorch · OpenCV · Open3D · COLMAP

- Built a quality inspection system that takes a video of a manufactured part on a turntable, reconstructs its 3D shape, and compares it to the CAD or STL model to find where it is out of tolerance; achieved sub-1mm accuracy using only a phone camera

Maritime Vessel Trajectory Prediction 2026
PyTorch · LSTM · FastAPI · PostgreSQL · MQTT

- A system that takes live ship position data from Finland's maritime API and predicts each vessel's location and heading 3, 6, and 12 minutes ahead; served via a web dashboard with a live map

SKILLS

Programming Python, R	Libraries PyTorch, TensorFlow, Pandas, NumPy, Scikit-Learn, OpenCV
AI / ML Deep learning, transformers, LSTMs, CNNs, GANs, anomaly detection, time series	Computer Vision Object detection (YOLOv8), 3D reconstruction (COLMAP, Open3D)
Data Engineering PostgreSQL, FastAPI, Kafka, Airflow, Snowflake, MQTT	Cloud & HPC AWS, Microsoft Azure, SLURM, CSC

LANGUAGES

Slovene Native	English C2 Proficient	German B2 Upper Intermediate	Finnish B1 Intermediate
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